

Trade Policy Uncertainty and Stock Return Predictability: A Tale of Two Periods ¹

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Abstract

This paper presents novel evidence that trade policy uncertainty (TPU) is a statistically and economically significant predictor of U.S. stock market returns in- and out-of-sample, and it provides incremental forecasting power beyond a host of traditional predictors and alternative uncertainty measures. The ability of TPU to predict market returns is contingent on its own state. Higher TPU is followed by higher subsequent returns only when TPU is high. Through a present-value decomposition, we attribute this predictability to TPU's ability to anticipate future cash flows rather than discount rates. We further identify behavioral effects: higher TPU precedes lower investor sentiment and lower disagreement, and subsequent upward correction following underpricing contributes to higher stock returns. Importantly, both fundamental and behavioral channels operate exclusively during the high-TPU regimes, explaining the observed asymmetric predictability.

Keywords: Trade policy uncertainty, out-of-sample forecast, conditional predictability, return decomposition, mispricing.

JEL Classification: C22, C53, E62, G12.

1 Introduction

On April 2, 2025, U.S. President Donald Trump unveiled a new set of tariff modifications encompassing three key dimensions: (1) a universal 10% tariff applicable to all trading economies; (2) a 25% import duty on automotive products entering the U.S. market; and (3) targeted tariffs ranging from 20% to 46% imposed on selected economies with substantial trade surpluses vis-à-vis the United States, including Vietnam (46%), China (34%), Japan (24%), the European Union (20%), etc. This policy shift triggered significant international concern and immediate sell-offs across equity markets. In the two trading days following the announcement (April 3-4, 2025), U.S. stock markets erased approximately \$6.5 trillion in value. Similar market volatility occurred during President Trump’s previous term (2017-2021), where his “America First” trade policies—particularly tensions with China—frequently generated substantial global equity fluctuations.

[Insert Figure 1 here]

The U.S. tariff policy adjustments, unquestionably, have been the most key source of trade policy uncertainty (TPU), which is usually defined as the unpredictability of market participants’ perception of the direction and strength of future trade policies. Figure 1 shows the news-based TPU index proposed by Caldara et al. (2020). A prominent feature is that the TPU index undergoes a significant increase before major geopolitical tensions, abrupt tariff adjustments, and unprecedented supply chain disruptions. Recent episodes, such as the U.S.-China tariff escalations in 2023, the European Union’s post-pandemic carbon border taxes, and the ripple effects of Russia’s invasion of Ukraine on commodity markets, underscore how trade policy shifts can destabilize global equity markets. Unsurprisingly, TPU shows the most remarkable increase since Trump was elected again in 2024, reaching levels more than one-and-a-half times the maximum uncertainty during his first presidency, which reflects the various tariffs that have been threatened and recently implemented.

Unlike general economic policy uncertainty or macroeconomic uncertainty, TPU directly impacts corporate supply chains, cost structures, and end-market demand—particularly for multinational enterprises. This distinct uncertainty may exert unique and pronounced effects on expected cash flows (Caldara et al., 2020), thereby potentially capturing shifts in trade-related risk premiums more effectively. Although existing literature extensively documents

TPU’s real effects on trade, credit spreads, and corporate investment (Caldara et al., 2020), its asset pricing implications remain largely unexplored. The recent convergence of abrupt tariff shifts and equity market declines raises critical questions: (1) does TPU contain robust predictive information about future market excess returns? (2) what characterizes its predictive ability? and (3) through which economic channels does it operate? This paper aims to address these questions. Cochrane (2008) stresses that the market risk premium has an important impact on asset pricing, risk management, and the real economy, and its predictability is one of most important issues in finance. To the best of our knowledge, the ability of TPU to predict market return in the time-series has not been evaluated before.

Forecasting stock returns has always been a central challenge, attracting sustained attention from both academics and practitioners over the past 40 years, and it shows no signs of subsiding.² The most common approach to examining predictability is based on in-sample fit. It is important to note, however, that one of the criticisms of time-series return predictability made in Inoue and Kilian (2005) and Welch and Goyal (2008) is that in-sample stock return predictability often fails to translate into reliable out-of-sample forecasts, likely due to parameter instability and model uncertainty in linear predictive specifications. In particular, Welch and Goyal (2008) and Goyal et al. (2024) show that a long list of financial and macroeconomic predictors (such as the dividend-price ratio, book-to-market ratio, or term spread) from the literature shown to contain significant in-sample predictive power cannot deliver better forecasts of market excess returns relative to a simple benchmark forecast based on the historical average in a real-time, out-of-sample forecasting experiment. Thus, whether investors with access only to available information can beneficently time the market remains questionable.

In this paper, we employ the TPU index developed by Caldara et al. (2020), constructed via automated text analysis of seven major newspapers (Boston Globe, Chicago Tribune, The Guardian, Los Angeles Times, New York Times, Wall Street Journal, and Washington Post). The index quantifies monthly TPU by measuring the frequency of trade policy-related articles (as a percentage of total news volume) for each outlet, normalized to 100 for a 1% article share. Our analysis uses the log-transformed TPU index, with sample period spanning from 1960 to 2024.

Our first major contribution to the literature is that we show for the first time that TPU

²See, for example, Atanasov et al. (2020), Pettenuzzo et al. (2020), Chen et al. (2022), Dong et al. (2022), Dong et al. (2022) Bao et al. (2023), Farmer et al. (2023), Yu and Huang (2023), Goyal et al. (2024), among others.

predicts stock returns at the market level, both in- and out-of sample. We consider the standard predictive regressions by regressing the S&P 500 market excess returns on the lagged TPU. The estimated slope coefficients are positive and statistically significant at conventional levels, confirming significant in-sample predictability. More critically, we find that the in-sample predictability retains out-of-sample. In our main results, we split the full sample into an initial training period spanning January 1960 to December 1989 and an evaluation period from January 1990 to December 2024. The out-of-sample R_{OS}^2 statistic of TPU overwhelmingly positive across horizons. We further conduct the mean squared forecasting error (MSFE)-adjusted tests of Clark and West (2007). The testing results show that TPU delivers significantly better forecasts of market excess returns relative to the historical average. Whether the superior out-of-sample forecasting performance of aggregate TPU can yield sizable economic gains for mean-variance investors? The answer is yes. We show that the TPU-based forecasts provide improved annualized gains in certainty equivalent return (CER) and Sharpe ratio relative to historical average returns for a mean-variance investor from an asset allocation perspective.

Our second major contribution is to document a state-dependent predictability pattern. While these full-sample regressions provide compelling evidence that TPU can explain variation in expected excess returns, we gain deeper insights into TPU’s predictive ability by splitting TPU into high and low periods. We classify month t as a high TPU period if the TPU index value falls into the upper tercile, and as a low TPU period otherwise. Conditional two-state forecasting regressions reveal a significant slope coefficient only during high-TPU periods. Out-of-sample exercises further show that TPU generates statistically significant forecasts beyond the historical average benchmark solely in the high-TPU states. For instance, the high-TPU out-of-sample R^2 ($R_{OS,high}^2$) reaches 26.9% at 24 months. In stark contrast, the low-TPU $R_{OS,low}^2$ statistics are consistently negative across all horizons. This asymmetric, state-dependent pattern also manifests economically. During the high-TPU periods, CER and SR gains are positive, indicating significant economic value. In contrast, during low-TPU periods, these gains become negligible or negative.

Our third notable contribution is to show that TPU delivers offer superior out-of-sample forecasting performance than the traditional predictors, and that it provides significant incremental predictive information beyond that is conveyed by those predictors. widely used in the literature is another natural question. We consider three sets of alternative predictor variables including the (1) commonly used financial valuation ratios, (2) macroeconomic indicators, and

(3) other types or measures of economic uncertainty. First of all, we re-produce the out-of-sample performance of TPU but replace the historical mean benchmark with return forecasts generated by a univariate predictive regression model using one of the alternative competing predictors. A notable finding is that the associated R_{OS}^2 are significantly positive, and therefore TPU offers superior out-of-sample forecasting performance than the traditional predictors. Second, we find bivariate weighted combination forecasts of returns using TPU plus one of the alternative predictors can offer superior out-of-sample forecasting performance relative to a benchmark that only incorporates the alternative predictors. Thus, TPU provides significant incremental predictive information complement to the traditional predictors.

To understand why TPU positively predicts future market returns and why the predictive ability is limited to high TPU periods only, we explore possible underlying economic mechanisms. First, according to the present-value framework of Campbell and Shiller (1988), TPU can influence stock returns through two fundamental channels: shocks to expected cash flows (dividend growth) and/or shocks to discount rates (expected returns). We perform a Campbell-Shiller-Vuolteenaho decomposition to split return innovations (i.e., unexpected returns) into cash flow and discount rate news components. Our results suggest that cash flow betas are statistically significant at conventional levels, but the significance of discount rate betas is only marginal. Thus, we conclude that the predictability of TPU mainly stem from its ability to predict future expected dividend growth rates, consistent with the finding in Caldara et al. (2020) that TPU has significant effect on investment at both firm and aggregate levels. Moreover, we isolate the relative importance of the cash flow and discount rate news components further in high and low TPU states. We find that TPU’s predictive ability during the high-TPU periods is likewise driven by the cash flow channel, but both channels contribute minimally to variation in expected future returns during the low-TPU periods.

Next, we provide behavior-based explanations. A well-established literature examines how economic agent’s subjective behaviors, like sentiment, disagreement, and attention, affect aggregate stock returns in the cross-section and time-series (see, e.g., Baker and Wurgler (2006, 2007) and Huang et al. (2015)). But how exactly these behavior transmission variables affect the predictive ability of TPU? We use investor sentiment to illustrate our story. When TPU rises, investors become more pessimistic because of the additional trade-policy-specific fear or uncertainty induced by TPU, and hence sentiment is lower. During the low-sentiment periods, the mispricing mechanism leads to current prices lower than their fundamentals, and the

upward correction of under-pricing will lead to higher stock returns in the future, according to Huang et al. (2015). Our empirical tests show that higher TPU precedes higher investor sentiment, higher attention, and lower investor disagreement, which are consistent with our story. To understand the asymmetric predictability through the behavior channel, we compare the predictive power of TPU for the behavior variables across high and low TPU states. The results, again, indicate that TPU is significantly associated with investor behavior measures during the high-TPU periods only. Thus, we roughly conclude that investors care about TPU only when this index is high enough.

Lastly, we tie the predictability of TPU to the real economy. Asset returns vary with business cycle in a manner than stock return predictability is primarily confined to economic recessions (see, for example, Fama and French (1989), Lettau and Ludvigson (2001), and Cochrane (2017)). If stock return forecasts are indeed related to macroeconomic fluctuations, return predictors should also forecast business cycles. The overall picture is that TPU is a significant negative predictor of economic activities in the near future, including GDP growth, industrial production, and personal consumption expenditure. However, we find different predictability patterns across TPU states. When TPU is high, the predictive power of TPU for economic activities is negative and significant at 5% up to 24 month, and the associated $NW-t$ statistics are greater than those obtained by the full-sample in Panel A. In contrast, when TPU is low, none of the three business cycle variables are predictable by TPU. Therefore, the documented correlations between TPU and business cycles further explains the asymmetric ability of TPU to predict stock returns across different TPU states.

The remainder of this paper is organized as follows. Section 2 describes the data used in this paper. In Section 3, we provide main in-sample regression results, out-of-sample forecasting results, and asset allocation exercises. In Section 4, we conduct an extensive set of out-of-sample tests compare the forecasting performance of TPU with the alternative popularly used predictor variables that come from the extant literature. In Section 5, we explore several possible underlying economic mechanisms to understand why TPU positively predicts future market returns and why the predictive ability is limited to high TPU periods only. Section 6 remarkably concludes.

2 Data

2.1 TPU

We consider the TPU index proposed by Caldara et al. (2020). The TPU data are provided by Matteo Iacoviello’s website.³ Simply speaking, this index is based on automated text searches of the electronic archives of seven newspapers: Boston Globe, Chicago Tribune, Guardian, Los Angeles Times, New York Times, Wall Street Journal, and Washington Post. The measure is calculated by counting the monthly frequency of articles discussing trade policy uncertainty (as a share of the total number of news articles) for each newspaper. The index is then normalized to a value of 100 for a one percent article share. The TPU Index starts in 1960.

2.2 Stock return

The aggregate stock market return is the continuously compounded log return on the S&P 500 index (including dividends). The excess returns are computed by subtracting the return on the 30-day T-bill from the raw return.

2.3 Competing predictor variables

To align our findings with the broader literature on stock return prediction, we consider three sets of alternative predictor variables.

The first set is the commonly used financial valuation variables, including (1) the log dividend-price ratio (DP), which is defined as the log of the 12-month moving average sum of dividends paid on the S&P 500 index minus the log of stock prices, (2) the log dividend yield (DY), which is defined as the log of the 12-month moving average sum of dividends paid on the S&P 500 index minus the lagged log of stock prices, (3) the log earning-price ratio (EP), which is defined as the difference between the log of earnings on the S&P 500 index and the log of prices, (4) the log dividend-payout ratio (DE), which is defined as =the log of the 12-month moving sum of dividends minus the log of stock prices, (5) the log book-to-market ratio (BM), which is defined as the log of the book-to-market value ratio for the Dow Jones Industrial Average (DJIA), and (6) the net equity expansion (NTIS), which is defined as the ratio of the 12-month moving sum of net equity issues by NYSE-listed stocks to the total end-of-year mar-

³<https://www.matteoiacoviello.com/tpu.htm>

ket capitalization of NYSE stocks. These data are obtain from Amit Goyal’s personal website.⁴ (7) excess stock return volatility (RVOL) is computed using the 12-month moving standard deviation estimator, as in Mele (2007).

The second set is the interest-rate-related macroeconomic variables, including (1) T-bill rate (TBL), which is defined as the interest rate on a three-month Treasury bill (secondary market), (2) long-term yield (LTY), which is defined as the long-term government bond yield, (3) long-term return on the long-term government bonds (LTR), (4) default yield spread (DFY) is the difference between Moodys BAA- and AAA-rated corporate bond yields, (5) default return spread (DFR) is the long-term corporate bond return minus the long-term government bond return, and (6) inflation (INFL) is the Consumer Price Index (All Urban Consumers) from the Bureau of Labor Statistics. The data are obtained from the Federal Reserve Economic Data(FRED).⁵ (7) term spread (TMS) is the difference between long-term yield and the Treasury bill rate.

The last set is the other types of economic uncertainty measures, including the 1-month and 3month ahead financial uncertainty of Ludvigson et al. (2021) (FU1 & FU3), the 1-month and 3month ahead macroeconomic uncertainty of Jurado et al. (2015) (MU1 & MU3), the geopolitical risk index (GPR) of Caldara and Iacoviello (2022), the variance risk premium (VRP) of Bollerslev et al. (2009), and the professional forecasters uncertainty (PFU) of Rossi and Sekhposyan (2015).

3 Forecasting Performance

3.1 Preliminary regression results

We use a standard predictive regression model to analyze aggregate stock return predictability, taking the form:

$$r_{t+1,t+h} = \alpha_h + \beta_h TPU_t + \epsilon_{t+1,t+h}, \quad (3.1)$$

where $r_{t+1,t+h} = (r_{t+1} + \dots + r_{t+h})/h$ denotes the annualized h -month ahead cumulative excess market returns in months $t + 1$ through $t + h$, and TPU_t is the log TPU index known at time t . Following Rapach et al. (2016), we standardize TPU_t to have a zero mean and unit standard deviation. Note that in the long-horizon regression (3.1), the dependent variable ($r_{t+1,t+h}$) is

⁴<https://sites.google.com/view/agoyal145/home>

⁵<https://fred.stlouisfed.org/>

sum of the single-period stock return over h horizons, and the use of overlapping observation may lead to the issue of serial correlation (see, for example, Campbell and Yogo (2006) and Ang and Bekaert (2007)). To test the significance of the long-horizon coefficient β_h , following Rapach et al. (2016), Atanasov et al. (2020), Yu and Huang (2023), among others, we use the heteroskedasticity- and autocorrelation-robust t -statistic of Newey and West (1987) (NW- t , hereafter) truncated at lag h .

Moreover, to test whether the ability of TPU to predict market excess returns is contingent on itself, we extend the above single-regime predictive regression (3.1) to conditional predictive regression models in the spirit of Boyd et al. (2005) and Atanasov et al. (2020) by allowing the predictive relation to switch across TPU regimes (high and low). In specific, we consider the following two-state predictive regression:

$$r_{t+1,t+h} = \alpha_h + \beta_{\text{high},h} S_t^{\text{high}} \text{TPU}_t + \beta_{\text{low},h} S_t^{\text{low}} \text{TPU}_t + \epsilon_{t+1,t+h}, \quad (3.2)$$

where S_t^{high} (S_t^{low}) is a dummy variable that equals one if month t is in high (low) TPU states. We define high-TPU states as periods in which the TPU index falls in its upper tercile. In the model, the slope coefficients $\beta_{\text{high},h}$ and $\beta_{\text{low},h}$ capture the degree of the predictive ability of TPU for h -month ahead cumulative excess market returns in high- and low-TPU periods, respectively.

[Place Table 1 about here]

Table 1 reports the in-sample regression results of stock returns for forecasting horizons ranging from one month to 36 months, covering the sample period from January 1960 to December 2024. Following the predictive regression model specified in equation (3.1), we first assess the overall in-sample predictive power of TPU over the full sample. Panel A reports the results. The results demonstrate a statistically significant predictive effect of TPU on future excess stock market returns. For example, at the one-month horizon, the estimated coefficient on TPU is 0.23, which is statistically significant at the 5% level as indicated by the NW t -statistic. Importantly, we find that the predictive power of TPU tends to be stronger at longer horizons. While at the six-month horizon, the t -statistic of the slope coefficient is 2.25 and the associated R^2 is only 1.53%, at horizons exceeding 24 months, the t -statistics consistently exceed three and the R^2 values rise above 5%.

Panel B summarizes the differences in the in-sample predictive ability of TPU between the high and low TPU regimes. The overall picture is that TPU performs well during the high-TPU periods, but its predictive strength is attenuated when TPU itself is low. In specific, during the high-TPU periods, β_{high} is statistically significant at 10% over one to six months, at 5% over 12 to 18 months, and at 1% over 24 to 36 months, much higher than those obtained by using the full sample. In terms of fit, the high-TPU regime's explanatory power is economically meaningful, with segmented R^2 statistics ranging from 0.81% to 20.88% — a notably high predictive performance within the stock return prediction literature. In stark contrast, the low-TPU coefficient β_{low} achieves statistical significance only at the 12-, 18-, and 24-month horizons. This disparity demonstrates that TPU's predictive power is markedly stronger during high-TPU regimes than during low-TPU regimes.

In summary, the results suggest that while TPU serves as a robust predictor of stock returns over the full sample, its predictive ability is predominantly concentrated in high-TPU periods only. When TPU levels are low, this TPU index largely loses its ability to forecast returns.

3.2 Out-of-sample results

One of the criticisms of time-series return predictability made in Welch and Goyal (2008) is that in-sample stock return predictability often fails to translate into reliable out-of-sample forecasts, likely due to parameter instability and model uncertainty in linear predictive specifications.

To assess out-of-sample stock return predictability, we calculate an out-of-sample R^2 , denoted as R_{OS}^2 , as in Campbell and Thompson (2008), Welch and Goyal (2008), and Van Binsbergen and Koijen (2010):

$$R_{OS}^2 = 1 - \frac{MSFE_P}{MSFE_B}, \quad (3.3)$$

where $MSFE_P = \frac{1}{T_2} \sum_{s=1}^{T_2} (r_{T_1+s} - \hat{r}_{T_1+s})^2$ is the mean-squared forecast error (MSFE) based on the fitted value from the predictive regression model (3.1) using TPU or one of the alternative predictor variables, $MSFE_B = \frac{1}{T_2} \sum_{s=1}^{T_2} (r_{T_1+s} - \bar{r}_{T_1+s})^2$ is the benchmark MSFE based on the historical average return estimated through period t (implying constant expected returns), T_1 denotes the length of the initial in-sample estimation period, and the rest $T_2 = T - T_1$ observations serve as the out-of-sample evaluation period. Following Welch and Goyal (2008), we consider a recursive estimation window in which each model is first estimated based on the first T_1 data and then recursively estimated by adding one month at a time until the end of

our sample. To investigate whether the model-based forecast has a significantly lower MSFE than the historical average benchmark forecast ($MSFE_P < MSFE_B$), implying $R_{OS}^2 \leq 0$, we implement the MSFE-adjusted test statistic proposed by Clark and West (2007). The formal hypotheses are $H_0 : R_{OS}^2 \leq 0$ vs. $H_A : R_{OS}^2 > 0$.

[Place Table 2 about here]

Table 2 presents the results of out-of-sample forecasts, with an initial training period spanning January 1960 to December 1989 and an evaluation period from January 1990 to December 2024. At the one-month horizon, TPU exhibits weak predictive power, yielding a negative out-of-sample R_{OS}^2 of -0.24 and a CW-test statistic of 0.55. Thus, TPU fails to outperform the historical average benchmark in a statistically significant manner. At longer horizons, however, TPU demonstrates significant out-of-sample predictive power. The associated R_{OS}^2 statistics increase from 0.6% (6-month) to 5.75% (18-month), and the associated Clark-West test statistics consistently exceed 2.5 (significant at the 1% level), indicating that TPU generates significantly lower MSFEs than the historical average at intermediate-to-long horizons.

Next, we look at the out-of-sample forecasting performance of TPU in the high- and low-TPU periods, respectively. Results are displayed in the last four rows of Table 2. In line with the evidence from the full-sample forecasting results, TPU fails to produce statistically significant forecasts beyond the historical average during the high-TPU periods at the one-month horizon, despite a marginally positive $R_{OS,high}^2$. Turning to long horizons, TPU shows impressive predictive power. For example, the high-TPU $R_{OS,high}^2$ value is 22.89% at the 18-month horizon and 26.85% at the 24-month horizon, much higher than the full-sample R_{OS}^2 statistics. The calculated CW-test statistics also illustrate that $R_{OS,high}^2$ is statistically higher than zero at the 1% level. Conversely, as shown in the last two rows, the $R_{OS,low}^2$ statistics are overwhelmingly negative across all forecasting horizons, accompanied by negligible CW-test statistics. This indicates TPU's MSFE systematically exceeds that of the historical average benchmark at conventional significance levels during the low-TPU periods.

Overall, the results from Table 2 establish a state-dependent return prediction pattern. When TPU is high, future stock returns are very sensitive to changes in current TPU; conversely, when TPU is low, future returns are not responsive to changes in current TPU.

3.3 Market-timing investment strategies

We have established that TPU significantly predicts the market returns from one month up to four years into the future, using both in-sample and out-of-sample tests. Now we tend to evaluate the economic significance associated with the out-of-sample predictive power of TPU for the excess market return.

Consider a mean-variance investor who decides to optimally allocate their wealth between the equity market portfolio and a risk-free bill at each time t . At the end of period t , the investor optimally allocates portfolio

$$w_t = \frac{1}{\gamma} \frac{\hat{r}_{t+1}}{\hat{\sigma}_{t+1}^2},$$

of the portfolio to equities during period $t + 1$, where γ is the risk-aversion parameter, $\hat{r}_{t+1}$ is the predicted value of monthly excess return on the aggregate stock market based on the predictive regression model, and $\hat{\sigma}_{t+1}^2$ is the variance forecast. Our economic evaluation settings are standard. Following Campbell and Thompson (2008), we assume that the investor uses a five-year moving window of past monthly returns to estimate the variance of the excess market return. In our baseline results, the risk-aversion γ is set to be three, following Rapach et al. (2016). We restrict the weight w_t to lie between 0 and 1.5 to exclude short sales and to allow for at most 50% leverage.

Following Campbell and Thompson (2008), Huang et al. (2015), among others, we compute the certainty equivalent return (CER) gain and Sharpe Ratio for the portfolio performance. Specifically, the investor realizes an average CER of the portfolio throughout the out-of-sample evaluation period, calculated as: $CER(\mu, \sigma) = \hat{\mu}_p - 0.5\gamma\hat{\sigma}_p^2$, where $\hat{\mu}_p$ and $\hat{\sigma}_p^2$ are the sample mean and variance, respectively, for the investment portfolio over the whole out-of-sample evaluation period. We compute the CER gain as the differences in the realized CER between investors who use TPU and those who use the sample mean of historical returns as the forecasts. We annualize the CER gains to provide the units of expected annualized return, which can be interpreted as the transaction costs or portfolio management fees for mean-variance investors who would be willing to pay to have access to the predictive regression instead of the historical mean forecast. We assume that the investor rebalances at the same frequency as the forecast horizon to analyze the economic value of predictability over the long horizons. Moreover, we also calculate the monthly Sharpe ratio (SR) of the portfolio, which is the mean of the excess

portfolio return throughout the out-of-sample evaluation period (in excess of the risk-free rate) divided by the standard deviation of the excess portfolio return. We compute the SR gain as the differences in the realized SR between investors who use TPU and those who use the sample mean of historical returns.

[Place Table 3 about here]

Table 3 reports the portfolio performance measures including CER gains and SR for a mean-variance investor with a risk aversion coefficient of 3 who allocates between equities and risk-free bills using the out-of-sample TPU-based forecast relative to the prevailing mean benchmark forecast. The out-of-sample evaluation period is from January 1990 to December 2024, consistent with the previous out-of-sample forecasting exercises. Over the full sample, the CER gains for TPU are all positive. The SR of portfolios formed based on TPU is higher than those formed based on the historical average of returns across, except for the one-month horizon, which is consistent with the small R_{OS}^2 statistics in Table 2. During the high-TPU periods, all CER and SR gains are positive, indicating significant economic value. In contrast, during the low-TPU periods, these gains are either negligible or negative. These results suggest that TPU produces economically meaningful out-of-sample return forecasts that help mean-variance investors to time the market.

4 Comparison with Traditional Predictors

Over recent decades, numerous financial, macroeconomic, and uncertainty measures have been used to predict stock returns. In this section, we compare the predictive information contained in TPU with that of these traditional predictors drawn from the extant literature.

4.1 Methodologies

Specifically, we address two key research questions: (1) Does TPU exhibit stronger predictive power than the traditional variables? and (2) Does TPU provide incremental out-of-sample predictive information beyond the traditional predictors? To address our questions, we employ two methods to compare the predictive power of TPU with traditional variables. First, we employ two univariate predictive regression models to compare the forecasting performance of

TPU against traditional predictors:

$$r_{t+1,t+h}^{(1)} = \alpha_J^{(1)} + \beta_J^{(1)} TPU_t + \varepsilon_{t+1,t+h}^{(1)}, \quad (4.1)$$

$$r_{t+1,t+h}^{(2)} = \alpha_J^{(2)} + \beta_J^{(2)} z_t + \varepsilon_{t+1,t+h}^{(2)}, \quad (4.2)$$

where TPU_t denote the log TPU index at time t and z_t denotes one of the alternative competing variables. Here, we test whether model (4.1) exhibits superior out-of-sample forecasting performance relative to model (4.2).

Second, we implement a pairwise combination forecast approach to assess whether TPU offers incremental predictive information beyond traditional variables. Following Rapach et al. (2010), we construct the bivariate combination forecast for the long-horizon stock return $r_{t+1,t+h}$ made at time t as a weighted average of the two individual forecasts using TPU and one of the alternative predictors:

$$\hat{r}_{t+1,t+h}^{(c)} = \omega_1 \hat{r}_{t+1,t+h}^{(1)} + \omega_2 \hat{r}_{t+1,t+h}^{(2)}, \quad (4.3)$$

where $\hat{r}_{t+1,t+h}^{(1)}$ and $\hat{r}_{t+1,t+h}^{(2)}$ are the predicted returns from models (4.1) and (4.2), respectively. Our focus is to test whether the combination forecast (4.3) could deliver significant out-of-sample forecasting performance relative to the univariate model (4.2). Throughout our analysis, we employ equal weights ($\omega_1 = \omega_2$) for the primary combination forecasts. Robustness checks using the discounted mean squared forecast error (DMSFE) weighting approach yield even stronger results.

4.2 Comparing with financial variables

The rational no-bubble present-value model of Campbell and Shiller (1988) posits that variation in the current dividend-price ratio is related to changes in the expected discount rates and/or expected cash flows. Therefore, the dividend-price ratio, earnings-price ratio, and a number of other price-scaled valuation ratios have been considered as regular predictors of stock returns. The predictive power of these financial indicators is well-documented in asset pricing literature (Fama and French, 1988; Campbell and Shiller, 1988). In this subsection, we aim to whether TPU subsumes or supplements predictive information contained in the traditional financial predictors. The financial predictors we consider include DP, DY, EP, DE, BM, NTIS, and

RVOL, which are detailed in Section 2.3.

[Place Table 4 about here]

Table 4 presents the comparison results. We first examine TPU’s out-of-sample forecasting performance relative to traditional financial predictors (i.e., TPU vs financial predictor). Importantly, most $R_{OS,UV}^2$ statistics, which measure the percentage MSFE reduction from using TPU alone versus a benchmark model using only traditional predictors, are positive across horizons. At the one-month, TPU outperforms DP, DY, DE, and NTIS in the full sample. For horizons beyond 12 months, TPU generates significantly more accurate forecasts than all of the traditional financial predictors, with gains statistically significant at the 1% level. During the high-TPU periods, the $R_{OS,UV}^2$ statistics remain consistently positive. For example, at the 24-month horizon, the $R_{OS,UV}^2$ for TPU relative to DP is 46.12%, meaning that TPU can deliver about 50% percentage gain in terms of out-of-sample forecasting accuracy relative to DP under these conditions. These results provide strong evidence that TPU exhibits significantly stronger predictive power than traditional financial predictors.

Next, we analyze whether a combination of TPU and traditional financial predictors delivers better out-of-sample forecasting performance relative to its benchmark with only the traditional financial predictors (i.e., TPU vs TPU+financial predictor). In Table 4, we construct $R_{OS, COMB}^2$ as the percent reduction in MSFE for the weighted combination forecasts relative to the forecasts based on the traditional financial variable alone. We find that the $R_{OS, COMB}^2$ statistics are almost positive across different horizons and TPU states. The CW-test statistics, which we omit in the table for simplicity, indicate that there is a statistically significant improvement in terms of forecasting accuracy when combining TPU to the model with only the financial predictors. These results demonstrate that TPU also contains sizable forecasting information complementary to what is contained in the predictors.

4.3 Comparing with macroeconomic variables

Macro-finance theory suggests that predictable variations in stock returns often reflect how economic agents rationally respond to investment opportunities over time. Such variations are mainly driven by the time-varying risk aversion that may be linked to the observed state of the business cycle—high in bad time and low in good time. Thus, in addition to financial variables, macroeconomic indicators are also commonly used to predict stock returns. We therefore

proceed to compare the predictive power of the TPU index with that of key macroeconomic indicators: DFR, DFY, INFL, LTR, LTY, TBL, and TMS, which are detailed in Section 2.3.

[Place Table 5 about here]

The results are presented in Table 5. First, we first compare forecasts using the TPU index individually against those based on traditional macroeconomic predictors. At the one-month horizon, forecasting performance is mixed. TPU show stronger predictive power than DFY and TMS at 5% and INFL and LTR at 10%, but does not surpass DFR, LTY, or TBL in a statistical significance way. Forecasting improvements for TPU tend to be more pronounced as the horizon extends. For example, at the 24-month horizon, TPU substantially outperforms DFR with a $R^2_{OS,UV}$ statistic of 6.65% and DFY with a $R^2_{OS,UV}$ statistic of 8.31%. However, TPU is unable to outperform the well-established long-term predictors TBL and TMS, as indicated by the negative $R^2_{OS,UV}$ statistics. Strikingly, TPU’s performance relative to TBL and TMS improves significantly, producing $R^2_{OS,UV}$ statistics of 16.27% and 38.7%, respectively. Moreover, the combined forecasts, which integrate both TPU and a traditional macroeconomic predictor, generally outperform forecasts using traditional predictors alone. While short-term gains remain modest (e.g., 0.61% at $h = 1$ combined with DFY), the combination forecast approach becomes more powerful at longer horizons, particularly in high-TPU conditions. For instance, combining TPU with the TMS yields a 24.18% improvement at the 24-month horizon during the high-TPU periods.

The above results reinforce our core finding: TPU serves as a powerful predictor, particularly when TPU itself is at a high level, whose predictive information is not captured by the common macroeconomic predictors.

4.4 Comparing with other uncertainty measures

While results show TPU predicts stock returns, it’s unclear whether this reflects unique information or merely correlation with existing uncertainty measures (e.g., economic policy uncertainty, variance risk premium). Crucially, trade policy uncertainty—focusing on tariffs, trade rules, and regulatory barriers—creates distinct risks not fully captured by broader measures. of-sample significance of TPU is robust to these additions. In this subsection, we try to answer whether TPU provide novel insights beyond conventional uncertainty indicators, or is its predictability subsumed by established factors?

[Place Table 6 about here]

The results are presented in Table Table 6. At the one-month horizon, the $R_{OS,UV}^2$ statistics of TPU are almost uniformly negative, and thus it cannot outperform the traditional uncertainty measures. This reverses at longer horizons. For horizons ranging from 24 to 36 months, TPU consistently outperforms traditional uncertainty predictors across most forecast horizons, especially during high-TPU periods and very long horizons. For example, at the 12-month horizon, the $R_{OS,UV}^2$ statistics of TPU are positive (e.g., 21.17% for FU1, 21.52% for FU3, and 34.3% for MU1) and significant at the 1% level. This suggests that uncertainty regarding future tariffs, trade rules, and regulatory barriers appears more critical for explaining time-series variation in expected returns than general macroeconomic or monetary policy uncertainty, or alternative risk measures. Bivariate combination forecasts further confirm that TPU contains unique information beyond that captured by traditional uncertainty predictors, and that this information significantly enhances out-of-sample stock return prediction accuracy.

5 Understanding the Source of Predictability

5.1 Present-value decomposition of returns

Our analysis has established a significant contemporaneous relationship between TPU and future market excess returns. To understand the economic mechanism behind this positive association, we employ the present-value framework of Campbell and Shiller (1988). This framework suggests the observed increase in prices with rising TPU may be due to positive changes in current or expected future cash flows or increases in discount rates. In this subsection, we test whether the underlying economic channel of the ability of TPU to predict excess return is from cash flow news or discount rate news.

Campbell (1991) decomposes the unexpected stock return from t to $t + 1$ into two components:

$$\underbrace{r_{t+1} - E_t r_{t+1}}_{\eta_{t+1}^r} = \underbrace{(E_{t+1} - E_t) \sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j}}_{\eta_{t+1}^{CF}} - \underbrace{(E_{t+1} - E_t) \sum_{j=0}^{\infty} \rho^j r_{t+1+j}}_{\eta_{t+1}^{DR}}$$

where E_t denotes an expectation formed at the end of period t , r_t denotes the log stock return at time t , Δd_t denotes the log dividend growth at time t , and the parameter ρ is a number a little

smaller than one (0.97 for monthly data). The identity posits that the unexpected component of the stock return (η_{t+1}^r) may be due to news about future cash flows (η_{t+1}^{CF}) or news about future discount rates (η_{t+1}^{DR}), or both.

To decompose the relative contributions of discount rate and cash flow news, we use the VAR approach developed by Campbell (1991). Specifically, we consider a VAR model:

$$\mathbf{Y}_{t+1} = \mathbf{\Phi} \mathbf{Y}_t + \mathbf{u}_{t+1},$$

where $\mathbf{Y}_t = (r_t, dp_t, x_t)$, $\mathbf{\Phi}$ is known as the companion matrix of the VAR, x_t denotes an N -vector of predictors for $N \geq 1$, and $\mathbf{u}_{t+1}$ denotes innovations. The unexpected return, discount rate news, and cash flow news components can be expressed as

$$\begin{aligned}\eta_{t+1}^r &= \tilde{\mathbf{I}}_{N+2} u_{t+1}, \\ \eta_{t+1}^{DR} &= \tilde{\mathbf{I}}_{N+2} \rho \mathbf{\Phi} (I - \rho \mathbf{\Phi})^{-1} u_{t+1}, \\ \eta_{t+1}^{CF} &= \eta_{t+1}^r + \eta_{t+1}^{DR},\end{aligned}$$

where $\tilde{\mathbf{I}}_{N+2}$ denotes an $(N+2)$ -vector with one as its first element and zeros for the remaining elements. Following Campbell (1991), we directly model the discount rate news and back out the cash flow news as the residual. Therefore, $\hat{\eta}_{t+1}^r$, $\hat{\eta}_{t+1}^{DR}$, and $\hat{\eta}_{t+1}^{CF}$ are directly obtained by fitting the VAR model. By combining the above terms, the log stock return is then decomposed as: $r_{t+1} = E_t r_{t+1} + \eta_{t+1}^{CF} - \eta_{t+1}^{DR}$, and thus $\widehat{E}_t r_{t+1} = \hat{\eta}_{t+1}^r - \hat{\eta}_{t+1}^{CF} + \hat{\eta}_{t+1}^{DR}$.

To examine the source of the power of TPU to predict stock returns, we use the following regressions:

$$\widehat{E}_t r_{t+1} = \alpha_E + \beta_E TPU_t + e_{t+1}^E, \quad (5.1)$$

$$\hat{\eta}_{t+1}^{DR} = \beta_{DR} TPU_t + e_{t+1}^{DR}, \quad (5.2)$$

$$\hat{\eta}_{t+1}^{CF} = \beta_{CF} TPU_t + e_{t+1}^{CF}, \quad (5.3)$$

and obtain the estimated coefficients, $\hat{\beta}_E$, $\hat{\beta}_{DR}$, and $\hat{\beta}_{CF}$. The forecasting coefficient $\hat{\beta}$ estimated in the predictive regression, $r_{t+1} = \alpha + \beta TPU_t + e_{t+1}$, is decomposed as

$$\hat{\beta} = \hat{\beta}_E + \hat{\beta}_{CF} - \hat{\beta}_{DR}.$$

Our main focus is the significance of $\hat{\beta}_{DR}$ and $\hat{\beta}_{CF}$ to test the channel(s) of TPU's ability to predict future stock returns (discount rate and/or cash flow news).

[Place Table 7 about here]

Table 7 reports OLS estimates of β_{CF} and β_{DR} based on the full-sample predictive regression $Comp_{t+1} = \alpha_h + \beta TPU_t + \varepsilon_{t+1}$ in Panel A and the two-state predictive regression $Comp_{t+1} = \alpha + \beta_{high} S_t^{high} TPU_t + \beta_{low} S_t^{low} TPU_t + \varepsilon_{t+1}$ in Panel B, respectively, where $Comp_t$ denotes the estimated cash flow innovations ($\hat{\eta}_{t+1}^{CF}$) or discount rate innovations ($\hat{\eta}_{t+1}^{DR}$) of the S&P 500 log return for month t . The components are estimated using the Campbell (1991) VAR approach based on a VAR comprised of the variables in the first column, where r indicates the S&P 500 log return. Following Rapach et al. (2016), we take the set of the GW and uncertainty predictors as a proxy for market information, and therefore DP is always included in each VAR model.

In Panel A, when considering only stock returns and DP, the OLS estimate of β_{CF} is 0.21 with a t -statistic of 1.93 (significant at 5%), while the estimate of β_{DR} is only 0.01 with a t -statistic of 1.33 (significant at 10%). In comparison, $\hat{\beta}_{DR}$ contributes relatively little to the size of $\hat{\beta}$. When incorporating additional variables to proxy for market information, the estimates of β_{DR} are still negligible, again accounting for a relatively limited share of $\hat{\beta}$. In contrast, the estimate of $\hat{\beta}_{CF}$ remains stably at around two in magnitude and all are significant at the 5% level. These results indicate that TPU predicts future stock returns over the full sample, with this predictive ability primarily stemming from the cash flow channel, while the discount rate channel plays a secondary role. Similarly, when using the first principal component extracted from all economic predictors to proxy for the market information set, the conclusions keep unchanged.

In Panel B, we isolate the relative importance of the cash flow (β_{CF}) and discount rate news (β_{DR}) components in high and low TPU states. During the high-TPU periods, nearly all estimates $\hat{\beta}_{CF,high}$ are positive (and most of them statistically significant), and closely match the magnitude of the full-sample estimates $\hat{\beta}_{CF}$ in magnitude. Unsurprisingly, $\beta_{DR,high}$ is close to zero and insignificant in most cases. This indicates that TPU's predictive ability during the high-TPU periods is likewise driven by the cash flow channel. In contrast, in the low-TPU periods, both $\beta_{CF,low}$ and $\beta_{DR,low}$ are insignificant, implying that TPU contributes minimally to explaining variation in expected future returns within the present-value framework. Together,

these results clarify the asymmetric return predictability pattern observed in prior out-of-sample analysis that TPU possesses significant forecasting power for stock returns exclusively during the high-TPU periods.

5.2 Behavioral interpretations

Why does TPU predicts future stock returns? We offer behavior-based explanations in this section.

Investor sentiment. A well-established literature examines how economic agent’s subjective sentiment affects aggregate stock returns in the cross-section and time-series (see, for example, Baker and Wurgler (2006, 2007), Huang et al. (2015), Jiang et al. (2019), among others). Intuitively, sentiment refers to the overall attitude of investors toward future cash flows and investment risks. The prevailing finding is that high investor sentiment predicts strongly low returns due to the correction of overpricing over time. The mechanism is straightforward: when sentiment is high, investors exhibit excessive optimism, particularly toward stocks that are difficult to value objectively. These stocks are prone to overvaluation because their fundamentals, such as earnings, are uncertain, allowing speculative narratives to dominate pricing (mis-pricing). These overpriced stocks induced by overly optimism eventually correct and subsequent aggregate returns should be lower.

But how exactly does TPU affect sentiment? TPU captures unpredictability in trade policies (e.g., tariffs, trade wars), creating ambiguity about future corporate earnings and investment (Caldara et al., 2020), especially for firms dependent on global supply chains or international markets. Our story is that when TPU rises, it creates additional trade-policy-specific fear or uncertainty for investors, and hence it could be a driver of sentiment, which in turn affects stock returns. When TPU is higher, investors become more pessimistic because they fear future policies could hurt corporate earnings, especially for or sentiment-driven traders who may overreact to TPU. So, higher TPU leads to a decline in investor sentiment. During low-sentiment periods, the mispricing mechanism leads to current prices lower than their fundamentals, so that stocks are undervalued. The upward correction of under-pricing will lead to higher stock returns in the future, according to Huang et al. (2015). As a result, lower stock market returns are theoretically followed by higher TPU, which is indeed consistent with the positive sign found in our previous empirical finding. Conversely, when TPU decreases, then sentiment improves and leads to overvalued price, eventually leading to lower subsequent returns are due to the

downward correction.

Disagreement. Market disagreement is another possible explanation for TPU’s predictive ability. It is well documented in the literature that heightened investor disagreement leads to overpricing, which is a key driver of lower subsequent returns (see, for example, Yu (2011) and Carlin et al. (2014)). To be specific, when disagreement about a stock’s fundamental value increases, short-sale constraints prevent pessimistic investors from fully participating. This allows relatively more optimistic investors to dominate the market, driving the price above the fundamental value. Consequently, prices starting from this inflated level tend to correct downward more frequently and correction results in lower average ex post returns.

We hypothesize that TPU is negatively related to market disagreement. Our explanation is similar to Bloom (2009). When TPU is high, the ambiguity of the macro policy path becomes the core risk factor that dominates the market, and the predictive efficiency of corporate micro fundamentals is greatly reduced, forcing market participants to put aside differentiated analysis of company characteristics and turn to convergent conservative predictions of the policy shocks. Alternatively, the unpredictability of trade policy shocks causes the Bayesian learning mechanism to fail, and market participants find it difficult to calibrate the probability distribution of trade policy impacts through historical data, and then may choose to delay decisions or follow consensus, reducing the heterogeneity of opinion expression.

To empirically validate the behavior-based predictability channel, we address a key question: Does TPU forecast changes in investors’ sentiment, disagreement or attention? We estimate the following linear predictive regression model:

$$InvB_{t+1,t+h} = \alpha_h + \beta_h TPU_t + \epsilon_{t+1,t+h} \quad (5.4)$$

where $InvB_{t+1,t+h} = (r_{t+1} + \dots + r_{t+h})/h$ denotes the average investor behavior variable between months t and $t + h$, and TPU_t (standardized to have zero mean and unit standard deviation) is the trade policy uncertainty index in period t . To broaden our analysis, we also consider the investor disagreement index (DIS) of Huang et al. (2021) and the investor attention index (ATTN) of Da et al. (2025) in addition to the investor sentiment index (SENT) of Huang et al. (2015) as investor behavior measures. To further examine the potential asymmetry in TPU’s

predictive ability, we estimate the following two-state predictive regression model: regressions:

$$InvB_{t+1,t+h} = \alpha_h + \beta_{\text{high},h} S_t^{\text{high}} TPU_t + \beta_{\text{low},h} S_t^{\text{low}} TPU_t + \epsilon_{t+1,t+h}. \quad (5.5)$$

We compute NW t -statistics to assess the significance of the slope coefficients, correcting for potential heteroskedasticity and autocorrelation in the errors.

[Place Table 8 about here]

Table 8 presents formal tests. In Panel A, we find that TPU is negatively associated with future investor sentiment over short horizons no more than 12 months. For example, at the one-month horizon, the estimate of β is -0.18, significant at 1%, and at the 12-month horizon, the estimate of β is -0.12, significant at 5%. This significant inverse relationship aligns with our theoretical prediction. When TPU increases, investor sentiment drops due to their fear about future trade policy adjustments and prices are undervalued, eventually leading to lower subsequent returns driven by the upward price-correction. However, at either 24 or 36 months, the NW t -statistics are less than one in absolute value, and hence the forecasting coefficient is statistically insignificant at conventional levels. In turn, the sentiment-based explanation seems no longer valid at long horizons. This attenuation is theoretically anticipated. Because of the limits of arbitrage, mispricings induced by investor sentiment may persist briefly for several months, but they will be eliminated completely by arbitrageurs at a sufficiently long time. Specifically, Huang et al. (2015) document that the aligned investor sentiment index contains significant predictive power for stock returns within 12 months only. Consequently, we can attribute the short-horizon (less than one year) predictive power of TPU to sentiment transmission.

To further support the behavior-based explanation, we also consider investor disagreement and attention. We find that TPU is significantly and negative related to investor disagreement. The estimated coefficients are around -0.2, and the corresponding the NW adjusted t -statistic is consistently greater than 2.7 in absolute value from one-month to 24-month horizons (significance at 1%) and is -0.18 (significance at 5%). The results are consistent with our hypothesis—lower TPU causes higher investor disagreement, and in turn leads to overpricing, which drives lower subsequent returns. This result nicely match the recent tariff adjustments, which push an incredibly high TPU, and investors seem to agree on the negative impact of

theses adjustment (low disagreement), thereby leading to a panic selling of stocks and thus lower returns. Moreover, we look at the relationship of TPU with future investor attention. According to Da et al. (2011) and Da et al. (2025), investor attention is likely to be a necessary condition for generating sentiment and disagreement. From Table 8, we observe a significant positive relationship of TPU with future investor attention, i.e., higher TPU indeed forces investor pay more attention on the market. Thus, both disagreement and attention explain the predictive ability of TPU over short and long horizons, while sentiment holds for short horizons only.

In Panel B of Table 8, we compare the predictive power of TPU for the behavior variables across high and low TPU states. Overall, TPU shows significant predictive power for all these variables. For both disagreement and attention, β_{high} is statistically significant at all forecasting horizons and the NW- t statistics exceed the full-sample estimates reported in Panel A. For investor sentiment, β_{high} remains significant from one month to 12 months, again mirroring the full-sample results. Nonetheless, the NW- t statistics and R^2 are still higher than that of β reported in Panel A. In stark contrast, β_{low} remains statistically insignificant across all horizons. In summary, these results indicate that TPU predicts (i) higher investor sentiment, (ii) increased attention, and (iii) lower disagreement only during high-TPU periods. This state-dependent relationship explains the observed asymmetry in TPU’s predictive power for stock returns.

5.3 Link to the real economy

It is now widely accepted that asset returns vary with business cycle in a manner than stock return predictability is primarily confined to economic recessions. A possible explanation is that stock return predictability reflects investors’ rational responses to time-varying investment opportunities. They are driven by cyclical variation in risk aversion: investors’ willingness to bear risk is greater in economic expansions, but less in economic recessions (see, for example, Fama and French (1989), Lettau and Ludvigson (2001), and Cochrane (2017)). Consequently, if stock return forecasts are indeed related to macroeconomic fluctuations, return predictors should also forecast business cycles.

Thus, in section, we ask whether TPU also predicts business cycle variables. First, we

consider the following linear predictive regression model:

$$Macro_{t+1,t+h} = \alpha_h + \beta_h TPU_t + \epsilon_{t+1,t+h}, \quad (5.6)$$

where $Macro_{t+1,t+h} = (r_{t+1} + \dots + r_{t+h})/h$ denotes the average business cycle variable between months t and $t+h$, including the GDP growth (GDPG), industrial production growth (IPG), and personal consumption expenditure growth (PCEG). All these data are sourced from the Federal Reserve Economic Data (FRED) database. Second, to test further whether the ability of TPU to predict future economic activities is contingent on the state of TPU, we extend to a two-state conditional predictive regression model as follows:

$$Macro_{t+1,t+h} = \alpha_h + \beta_{\text{high},h} S_t^{\text{high}} TPU_t + \beta_{\text{low},h} S_t^{\text{low}} TPU_t + \epsilon_{t+1,t+h}, \quad (5.7)$$

Similarly, we compute NW t-statistics to assess the significance of the slope coefficients.

[Place Table 9 about here]

Table 9 presents the relationship between TPU and real economic activities. The overall picture in Panel A is that TPU is a significant negative predictor of economic activities in the near future. When predicting GDPG, the forecasting coefficient is -0.12 at the one-month horizon, suggesting a counter-cyclical pattern of TPU, consistent with Caldara et al. (2020), and the NW- t statistic is above the 1% threshold. However, between 12 months to 24 months, this coefficient remains statistical significance at 10% only, and it turns out to be completely insignificant at the longest horizon of 36 months. A similar pattern is observed when predicting TPG and PCEG. All results in Panel A reveal that TPU is only related to the near-term business cycles. In panel B, we document different predictability patterns across TPU states. When TPU is high, the coefficient $\beta_{\text{high},h}$ is negative and significant at 5% up to 24 month, and the associated NW- t statistics are greater than those obtained by the full-sample in Panel A. In contrast, when TPU is low, none of the three business cycle variables are predictable by TPU. In sum, the link between TPU and business cycle, further explains the asymmetric ability of TPU to predict stock returns across different TPU states.

6 Conclusion

This study provides robust and novel evidence that TPU serves as a statistically and economically significant predictor of US stock market excess returns, both in-sample and crucially, out-of-sample. The predictability exhibits a distinct state-dependent asymmetry: TPU positively forecasts future returns only during periods when uncertainty itself is elevated (i.e., upper-tercile TPU regimes). This conditional predictability holds across various horizons, and translates into substantial economic gains for mean-variance investors, as evidenced by significant improvements in certainty equivalent returns and Sharpe ratios during the high-TPU periods. Importantly, TPU offers significant complementary forecasting power beyond a wide array of traditional financial ratios, macroeconomic indicators, and alternative uncertainty measures.

Delving into the economic mechanisms, we attribute the source of TPU’s predictive power primarily to the cash flow channel. Through Campbell-Shiller present-value decomposition, we demonstrate that TPU forecasts future stock returns predominantly by anticipating changes in expected future cash flows (dividend growth), consistent with its documented negative impact on firm and aggregate investment. The discount rate channel plays a minimal role. Furthermore, behavioral factors provide an additional explanation: higher TPU leads to lower investor sentiment, reduced disagreement, and increased attention specifically during high-uncertainty regimes. This behavioral shift contributes to contemporaneous underpricing, whose subsequent correction drives the observed positive relationship between TPU and future returns. Both the fundamental (cash flow) and behavioral channels operate almost exclusively during high-TPU states, explaining the observed asymmetry in predictability.

Our work can be further extended in the following dimensions. Firstly, while our study establishes TPU as a state-contingent predictor of U.S. equity returns, we are yet to examine its predictive power extends to economies with higher trade openness or commodity-driven markets. Secondly, the industry heterogeneity in TPU exposure deserves attention. Sectors like manufacturing and technology, with complex global supply chains, may exhibit stronger cash-flow sensitivities than domestically oriented industries. Last but not the least, we can further investigate whether TPU commands a distinct risk premium in bond, currency, or derivatives markets. We leave this work for future research.

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Table 1: **In-sample predictive performance.** This table presents the estimated slope coefficient β (%), NW robust t -statistic (NW- t), and in-sample R^2 (%) of the predictive regression: $r_{t+1,t+h} = \alpha_h + \beta_h TPU_t + \epsilon_{t+1,t+h}$ (Panel A) or $r_{t+1,t+h} = \alpha_h + \beta_{\text{high},h} S_t^{\text{high}} TPU_t + \beta_{\text{low},h} S_t^{\text{low}} TPU_t + \epsilon_{t+1,t+h}$ (Panel B), where $r_{t+1,t+h} = (r_{t+1} + \dots + r_{t+h})/h$ denotes the average market return between months $t+1$ and $t+h$ for the S&P 500 index; TPU_t is the trade policy uncertainty index in period t ; S_t^{high} (S_t^{low}) is a dummy variable that equals one if month t falls in high (low) TPU periods where high and low periods are defined based on the upper tercile of the TPU index distribution. TPU is standardized to have a zero mean and unit standard deviation. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for the NW- t . The sample period is from January 1960 to December 2024.

	$h = 1$	$h = 6$	$h = 12$	$h = 18$	$h = 24$	$h = 30$	$h = 36$
Panel A: $r_{t+1,t+h} = \alpha_h + \beta_h TPU_t + \epsilon_{t+1,t+h}$							
β	0.23**	0.24**	0.26***	0.23***	0.21***	0.19***	0.18***
NW- t	1.66	2.25	2.70	2.98	3.25	3.35	3.11
R^2	0.28	1.53	3.42	4.14	4.83	5.32	5.51
Panel B: $r_{t+1,t+h} = \alpha_h + \beta_{\text{high},h} S_t^{\text{high}} TPU_t + \beta_{\text{low},h} S_t^{\text{low}} TPU_t + \epsilon_{t+1,t+h}$							
β_{high}	0.23*	0.17*	0.18**	0.18**	0.17***	0.18***	0.15***
NW- t	1.45	1.63	2.03	2.21	2.34	3.03	2.75
β_{low}	0.04	0.11	0.13*	0.10*	0.08*	0.05	0.06
NW- t	0.28	1.01	1.33	1.31	1.34	0.87	0.97
R^2	0.31	1.53	3.42	4.17	4.89	5.68	5.68
R_{high}^2	0.81	3.94	10.03	12.68	16.02	20.88	16.45
R_{low}^2	0.06	0.73	1.62	1.66	1.81	1.30	1.80

Table 2: **Out-of-sample predictive performance.** This table reports the out-of-sample performance of TPU in predicting the h -month ahead excess market return on the S&P 500 index. R_{OS}^2 is the Campbell and Thompson (2008) out-of-sample R^2 which measures the percent reduction in mean squared forecast error (MSFE) for the TPU-based forecast at h horizon relative to the historical mean benchmark forecast. CW-test is the Clark and West (2007) MSFE-adjusted statistic. The statistic test the null hypothesis that the historical mean MSFE is less than or equal to the TPU-based predictive regression MSFE, against the alternative hypothesis that the historical mean MSFE is greater than the TPU-based predictive regression MSFE. $R_{OS,high}^2$ ($R_{OS,low}^2$) statistic is calculated over high (low) TPU periods, where high TPU periods are defined as those in the upper tercile of the TPU distribution. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for the CW-test. The out-of-sample evaluation period is from January 1990 to December 2024.

	$h = 1$	$h = 6$	$h = 12$	$h = 18$	$h = 24$	$h = 30$	$h = 36$
R_{OS}^2	-0.24	0.60***	3.59***	5.50***	5.75***	4.52***	3.81***
CW-test	0.55	2.55	5.41	5.98	5.62	5.10	4.95
$R_{OS,high}^2$	0.47	6.61***	16.58***	22.89***	26.85***	21.81***	16.36***
CW-test	1.15	4.06	7.04	7.34	7.16	7.08	7.14
$R_{OS,low}^2$	-0.64	-1.53	-0.96	-0.48	-1.07	-1.40	-1.35
CW-test	-1.17	-1.33	-0.30	0.02	-0.84	-1.54	-1.23

Table 3: **Asset allocation results.** This table reports the portfolio performance measures for a mean-variance investor with a risk aversion coefficient of 3, who allocates between equities and risk-free bills using either the out-of-sample TPU-based return forecast or the historical average benchmark forecast. CER gain and SR gain refer to the annualized certainty equivalent return gain and Sharpe ratio gain, respectively, achieved by using the TPU-based forecast relative to the benchmark. The out-of-sample evaluation period is from January 1990 to December 2024.

	Full periods		High TPU periods		Low TPU periods	
	CER gain	SR gain	CER gain	SR gain	CER gain	SR gain
h=1	0.047	−0.023	1.732	0.025	−1.029	−0.150
h=6	0.431	0.045	2.344	0.129	−0.569	−0.064
h=12	0.395	0.047	2.757	0.145	−0.343	−0.015
h=18	0.562	0.058	1.475	0.086	0.255	0.049
h=24	0.280	0.027	1.585	0.073	0.021	0.020
h=30	0.173	0.012	1.791	0.135	−1.024	−0.093
h=36	0.456	0.028	2.975	0.160	0.035	−0.001

Table 4: **Comparison with financial variables.** This table reports the out-of-sample R^2 statistics (R^2_{OS}) for predicting h -month ahead excess market return on the S&P 500 index. $R^2_{OS,UV}$ measures the percent reduction in MSFE for the TPU-based forecast relative to the forecast based on one of traditional financial variables in the second column. $R^2_{OS,COMB}$ measures the percent reduction in MSFE for the weighted combination forecast of the individual forecasts from TPU and the traditional financial predictor, relative to the forecast based on the traditional financial variable alone. The financial variables include the log dividend-price ratio (DP), log dividend yield (DY), log earning-price ratio (EP), log dividend-payout ratio (DE), log book-to-market ratio (BM), net equity expansion (NTIS), and equity risk premium volatility (RVOL). Results are reported over the full periods, high TPU periods, and low TPU periods, where high and low periods are defined based on the upper tercile of the TPU index distribution. A positive R^2_{OS} indicates improved forecast accuracy relative to the corresponding benchmark. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for the CW-test. The out-of-sample evaluation period is from January 1990 to December 2024.

		$h = 1$			$h = 12$			$h = 24$			$h = 36$		
		Full	High	Low	Full	High	Low	Full	High	Low	Full	High	Low
$R^2_{OS,UV}$	DP	0.70**	0.89*	0.60*	17.82***	30.17***	13.39***	28.88***	46.12***	23.13***	27.97***	38.12***	23.74***
	DY	0.87**	1.30**	0.62*	17.55***	29.88***	13.14***	28.22***	45.16***	22.63***	27.51***	37.30***	23.47***
	EP	0.14	0.27	0.08	9.06***	18.59***	5.87***	12.62***	30.93***	6.85***	10.40***	22.18***	5.56***
	DE	1.03**	0.93*	1.08	7.76***	18.43***	4.13***	14.66***	31.48***	9.47***	12.57***	19.44***	9.97***
	BM	-0.12	0.14	-0.27	8.41***	19.39***	4.65***	14.08***	27.99***	10.02***	12.23***	14.36***	11.48***
	NTIS	1.17***	0.30*	1.65**	15.77***	15.38***	15.89***	11.39***	26.71***	6.84***	7.31***	21.70***	1.15**
	RVOL	-0.58	0.08*	-0.86	3.40***	15.49***	-0.77	9.57***	33.34***	1.35**	11.06***	22.94***	6.16***
$R^2_{OS,COMB}$	DP	0.59**	0.98*	0.38*	10.83***	20.26***	7.44***	15.94***	26.56***	12.39***	15.19***	20.73***	12.88***
	DY	0.70**	1.22**	0.40*	10.66***	20.08***	7.29***	15.53***	25.95***	12.10***	14.95***	20.27***	12.75***
	EP	0.24	0.49	0.10	5.75***	13.01***	3.32***	7.01***	17.78***	3.62***	5.61***	12.00***	2.99***
	DE	0.72**	0.85*	0.65	5.20***	12.76***	2.63***	9.42***	18.64***	6.57***	8.41***	11.29***	7.33***
	BM	0.09	0.39	-0.07	5.31***	12.98***	2.69***	7.91***	15.74***	5.62***	7.17***	7.91***	6.91***
	NTIS	0.87***	0.62*	1.01**	9.81***	12.49***	9.04***	6.77***	16.07***	4.01***	4.20***	11.91***	0.91**
	RVOL	0.09	0.61*	-0.21	3.47***	12.73***	0.28	5.80***	19.75***	0.98**	6.06***	12.64***	3.35***

Table 5: **Comparison with macroeconomic variables.** This table reports the out-of-sample R^2 statistics (R^2_{OS}) for predicting h -month ahead excess market return on the S&P 500 index. $R^2_{OS,UV}$ measures the percent reduction in MSFE for the TPU-based forecast relative to the forecast based on one of traditional macroeconomic variables in the second column. $R^2_{OS,COMB}$ measures the percent reduction in MSFE for the weighted combination forecast of the individual forecasts from TPU and the traditional macroeconomic predictor, relative to the forecast based on the traditional macroeconomic variable alone. The macroeconomic variables include the default yield spread (DFY), default return spread (DFR), inflation rate (INFL), long-term government bond return (LTR), long-term government bond yield (LTY), treasury bill rate (TBL), and term spread (TMS). Results are reported over the full periods, high TPU periods, and low TPU periods, where high and low periods are defined based on the upper tercile of the TPU index distribution. A positive R^2_{OS} indicates improved forecast accuracy relative to the corresponding benchmark. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for the CW-test. The out-of-sample evaluation period is from January 1990 to December 2024.

		$h = 1$			$h = 12$			$h = 24$			$h = 36$		
		Full	High	Low	Full	High	Low	Full	High	Low	Full	High	Low
$R^2_{OS,UV}$	DFR	0.51	1.49	-0.06	4.02***	15.78***	-0.02	6.65***	26.85***	0.21	4.03***	17.32***	-1.50
	DFY	0.76**	0.94**	0.66	4.44***	17.85***	-0.29	8.31***	29.43***	1.41**	8.19***	23.20***	1.68***
	INFL	0.36*	-0.28	0.72	-1.77**	4.91***	-3.88	2.43***	20.39***	-2.99	0.15**	9.49***	-3.46
	LTR	0.37*	1.73**	-0.42	4.06***	16.78***	-0.37	6.03***	26.27***	-0.41	3.64***	17.09***	-1.96
	LTY	-0.83	-0.78	-0.85	2.46***	8.98***	0.39	8.12***	21.62***	4.26***	8.48***	14.89***	6.09***
	TBL	-0.58	-0.40	-0.68	-0.99**	8.65***	-4.17	-0.50*	16.27***	-5.43	-3.63	5.82***	-7.27
	TMS	0.75**	2.78**	-0.44	4.82***	32.81***	-8.22	-3.05**	38.70***	-22.56	-13.68	24.25***	-36.90
$R^2_{OS,COMB}$	DFR	0.49	1.15	0.11	2.96***	11.04***	0.19	3.97***	15.64***	0.24	2.39***	9.70***	-0.66
	DFY	0.61**	0.96**	0.41	3.45***	12.80***	0.14	4.86***	17.23***	0.83**	4.53***	12.79***	0.95***
	INFL	0.36*	0.18	0.46	0.73**	6.20***	-0.99	2.26***	12.83***	-0.93	0.88**	5.86***	-1.05
	LTR	0.58*	1.44**	0.08	3.41***	12.02***	0.40	3.79***	15.56***	0.04	2.31***	9.69***	-0.76
	LTY	-0.29	0.21	-0.34	1.88***	6.65***	0.37	4.52***	12.35***	2.29***	4.62***	8.17***	3.29***
	TBL	-0.06	0.01	-0.10	0.53**	6.17***	-1.33	0.39*	9.46***	-2.28	-1.36	3.50***	-3.23
	TMS	0.75**	2.04**	-0.01	5.10***	21.77***	-2.67	1.17**	24.18***	-9.58	-3.73	16.04***	-15.83

Table 6: **Comparison with uncertainty variables.** This table reports the out-of-sample R^2 statistics (R^2_{OS}) for predicting h -month ahead excess market return on the S&P 500 index. $R^2_{OS,UV}$ measures the percent reduction in MSFE for the TPU-based forecast relative to the forecast based on one of traditional uncertainty variables in the second column. $R^2_{OS,COMB}$ measures the percent reduction in MSFE for the weighted combination forecast of the individual forecasts from TPU and the traditional uncertainty predictor, relative to the forecast based on the traditional uncertainty variable alone. The uncertainty variables include the 1-month and 3-month ahead financial uncertainty of Ludvigson et al. (2021) (FU1 & FU3), the 1-month and 3-month ahead macroeconomic uncertainty of Jurado et al. (2015) (MU1 & MU3), the geopolitical risk index (GPR) of Caldara and Iacoviello (2022), the variance risk premium (VRP) of Bollerslev et al. (2009), and the professional forecasters uncertainty (PFU) of Rossi and Sekhposyan (2015). Results are reported over the full periods, high TPU periods, and low TPU periods, where high and low periods are defined based on the upper tercile of the TPU index distribution. A positive R^2_{OS} indicates improved forecast accuracy relative to the corresponding benchmark. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for the CW-test. The out-of-sample evaluation period is from January 1990 to December 2024.

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		$h = 1$			$h = 12$			$h = 24$			$h = 36$		
		Full	High	Low	Full	High	Low	Full	High	Low	Full	High	Low
$R^2_{OS,UV}$	FU1	−0.99	1.72	−2.63	5.39***	21.37***	−0.42	12.55***	32.82***	6.19***	13.15***	19.59***	10.88***
	FU3	−1.01	1.72	−2.66	5.33***	21.52***	−0.57	12.54***	32.95***	6.11***	13.13***	19.72***	10.79***
	MU1	−0.27	3.27*	−2.43	8.24***	34.30***	−3.43	10.20***	36.70***	0.86*	9.03***	25.38***	2.22**
	MU3	−0.44	2.98*	−2.53	7.10***	32.88***	−4.29	9.03***	35.95***	−0.46	7.60***	24.36***	0.59
	GPR	−0.10	0.88*	−0.65	6.16***	17.66***	2.21***	11.48***	30.59***	5.40***	9.65***	22.59***	4.24***
	VRP	15.98	25.78	−11.63	15.13	26.88	−8.87	9.00**	54.78***	−39.67	−24.3	65.45***	−66.56
	PFU	−1.20	−0.57	−1.59	−1.55**	11.16***	−5.38	1.42**	27.70***	−4.42	−0.24*	22.88***	−5.95
$R^2_{OS,COMB}$	FU1	0.08	1.57	−0.83	4.47***	15.07***	0.62	7.56***	19.47***	3.81***	8.28***	11.81***	7.04***
	FU3	0.07	1.57	−0.84	4.47***	15.17***	0.57	7.57***	19.56***	3.79***	8.29***	11.90***	7.01***
	MU1	0.24	2.20*	−0.97	6.24***	21.58***	−0.63	7.00***	22.72***	1.46*	5.88***	15.26***	1.97**
	MU3	0.13	2.04*	−1.03	5.55***	20.78***	−1.18	6.22***	22.10***	0.62	4.98***	14.55***	0.98
	GPR	0.14	0.85*	−0.26	4.09***	12.06***	1.35***	6.60***	17.75***	3.04***	5.49***	12.55***	2.53***
	VRP	9.96	15.18	−4.78	10.29	16.95	−3.33	10.11**	35.80***	−17.20	−3.83	50.42***	−29.37
	PFU	−0.23	0.35	−0.59	1.38**	11.54***	−1.68	1.67**	16.42***	−1.61	0.73*	12.80***	−2.25

Table 7: **Cash flow and discount rate channels.** The table reports the β estimate and NW- t for the predictive regression: $Comp_{t+1} = \alpha_h + \beta TPU_t + \varepsilon_{t+1}$ (Panel A) or $Comp_{t+1} = \alpha + \beta_{\text{high}} S_t^{\text{high}} TPU_t + \beta_{\text{low}} S_t^{\text{low}} TPU_t + \varepsilon_{t+1}$ (Panel B), where $Comp_{t+1}$ denotes either the cash flow innovation ($\hat{\eta}_{t+1}^{\text{CF}}$) or the discount rate innovation ($\hat{\eta}_{t+1}^{\text{DR}}$) of the S&P 500 log return for month $t + 1$. The components are estimated using the vector autoregression (VAR) methodology of Campbell (1991), based on a VAR comprised of the variables in the first column, where r indicates the S&P 500 log return; PC indicates the first principal component extracted from all economic predictors in the first column. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for the NW- t . The sample period is from January 1960 to December 2024.

	Panel A: $Comp_{t+1} = \alpha + \beta TPU_t + \varepsilon_{t+1}$				Panel B: $Comp_{t+1} = \alpha + \beta_{\text{high}} S_t^{\text{high}} TPU_t + \beta_{\text{low}} S_t^{\text{low}} TPU_t + \varepsilon_{t+1}$							
	β_{CF}	NW- t	β_{DR}	NW- t	$\beta_{\text{CF,high}}$	NW- t	$\beta_{\text{CF,low}}$	NW- t	$\beta_{\text{DR,high}}$	NW- t	$\beta_{\text{DR,low}}$	NW- t
r, DP	0.21**	1.93	0.01*	1.33	0.20*	1.63	0.05	0.36	0.01	1.08	0.00	0.39
r, DP, DY	0.22**	1.97	0.01	0.73	0.20*	1.64	0.05	0.40	0.00	0.44	0.00	0.34
r, DP, EP	0.22**	1.98	0.01*	1.52	0.21**	1.68	0.04	0.35	0.01*	1.33	0.00	0.25
r, DP, DE	0.22**	1.94	0.00	0.03	0.20**	1.65	0.06	0.42	0.01	0.63	-0.01	-0.58
r, DP, RVOL	0.17*	1.58	0.04	0.74	0.21**	1.75	-0.01	-0.08	0.01	0.21	0.04	0.87
r, DP, BM	0.22**	1.98	0.02**	1.93	0.20*	1.62	0.05	0.42	0.01	0.80	0.02*	1.41
r, DP, NTIS	0.21**	1.92	0.00	-0.01	0.21**	1.69	0.03	0.27	0.00	0.01	0.00	-0.02
r, DP, TBL	0.22**	2.00	0.02	1.01	0.21**	1.71	0.04	0.30	0.03	1.25	-0.01	-0.45
r, DP, LTY	0.22**	1.96	0.02*	1.35	0.21*	1.64	0.04	0.30	0.01	0.78	0.01	0.67
r, DP, LTR	0.21**	1.92	0.02*	1.38	0.20*	1.58	0.04	0.35	0.01	0.32	0.02*	1.42
r, DP, TMS	0.22**	1.91	0.01	0.36	0.25**	1.89	0.00	-0.02	0.01	0.41	0.00	-0.02
r, DP, DFY	0.20**	1.88	0.02	1.11	0.19**	1.66	0.05	0.38	-0.02	-0.97	0.06***	2.34
r, DP, DFR	0.21**	1.90	0.01	1.13	0.20*	1.61	0.05	0.39	0.01	0.96	0.00	0.31
r, DP, INFL	0.22**	1.94	0.02*	1.40	0.20*	1.63	0.05	0.39	0.02*	1.51	0.00	0.20
r, DP, PC	0.22**	2.02	0.04*	1.62	0.21*	1.63	0.04	0.34	0.01	0.24	0.05**	1.83

Table 8: **Link to the investor behavior.** This table presents the estimated slope coefficient β (%), NW- t , and in-sample R^2 (%) of the predictive regression: $InvB_{t+1,t+h} = \alpha_h + \beta_{\text{high},h} S_t^{\text{high}} TPU_t + \beta_{\text{low},h} S_t^{\text{low}} TPU_t + \epsilon_{t+1,t+h}$, where $InvB_{t+1,t+h} = (r_{t+1} + \dots + r_{t+h})/h$ denotes the average investor behavior variable between months $t+1$ and $t+h$, including the investor disagreement index (DIS) of Huang et al. (2021), the investor attention index (ATTN) of Da et al. (2025), and the investor sentiment index (SENT) of Huang et al. (2015); TPU_t is the trade policy uncertainty index in period t ; S_t^{high} (S_t^{low}) is a dummy variable that equals one if month t falls in high (low) TPU periods where high and low periods are defined based on the upper tercile of the TPU index distribution. TPU_t is standardized to have a zero mean and unit standard deviation. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for the NW- t . The sample period is from January 1960 to December 2024.

	DIS				ATTN				SENT			
	$h = 1$	$h = 12$	$h = 24$	$h = 36$	$h = 1$	$h = 12$	$h = 24$	$h = 36$	$h = 1$	$h = 12$	$h = 24$	$h = 36$
Panel A: $InvB_{t+1,t+h} = \alpha_h + \beta_h TPU_t + \epsilon_{t+1,t+h}$												
β	-0.2 ***	-0.21 ***	-0.22 ***	-0.18 **	0.23 ***	0.27 **	0.44 ***	0.28 **	-0.16 ***	-0.13 *	-0.09	-0.09
NW- t	-5.13	-2.94	-2.71	-1.72	3.40	2.23	4.61	1.92	-3.66	-1.46	-0.91	-0.97
R^2	4.02	5.62	7.40	4.31	5.22	24.55	32.89	9.89	2.47	1.96	1.06	1.27
Panel B: $InvB_{t+1,t+h} = \alpha_h + \beta_{\text{high},h} S_t^{\text{high}} TPU_t + \beta_{\text{low},h} S_t^{\text{low}} TPU_t + \epsilon_{t+1,t+h}$												
β_{high}	-0.26 ***	-0.22 ***	-0.22 ***	-0.24 **	0.16 **	0.18 *	0.53 ***	0.54 ***	-0.14 ***	-0.11 **	-0.08	-0.04
NW- t	-7.45	-4.00	-3.38	-1.92	2.04	1.39	3.88	3.57	-5.14	-2.20	-1.25	-0.56
β_{low}	0.05	-0.01	-0.04	-0.02	0.11 *	0.13 *	0.06	0.01	-0.05	-0.04	-0.02	-0.07
NW- t	1.02	-0.13	-0.65	-0.25	1.51	1.53	0.74	0.18	-0.70	-0.32	-0.15	-0.59
R^2	5.77	6.43	7.91	5.36	5.29	25.20	37.17	16.51	2.57	2.02	1.13	1.44
R^2_{high}	14.05	18.80	21.70	14.29	10.00	29.35	62.22	40.68	10.75	7.55	3.48	2.12
R^2_{low}	-0.56	0.46	1.42	0.07	2.43	16.33	-2.69	-7.71	0.67	0.58	0.24	0.97

Table 9: Link to the real economy. This table presents the estimated slope coefficient β (%), NW- t , and in-sample R^2 (%) of the predictive regression: $Macro_{t+1,t+h} = \alpha_h + \beta_{\text{high},h} S_t^{\text{high}} TPU_t + \beta_{\text{low},h} S_t^{\text{low}} TPU_t + \epsilon_{t+1,t+h}$, where $Macro_{t+1,t+h} = (Macro_{t+1} + \dots + Macro_{t+h})/h$ denotes the average economic condition variable between months $t+1$ and $t+h$, including the gross domestic product growth (GDPG), the industry production growth (IPG), and the personal consumption expenditures growth (PCEG); TPU_t is the trade policy uncertainty index in period t ; S_t^{high} (S_t^{low}) is a dummy variable that equals one if month t falls in high (low) TPU periods where high and low periods are defined based on the upper tercile of the TPU index distribution. TPU_t is standardized to have a zero mean and unit standard deviation. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively, for the NW- t . The sample period is from January 1960 to December 2024.

	GDPG				IPG				PCEG			
	$h = 1$	$h = 12$	$h = 24$	$h = 36$	$h = 1$	$h = 12$	$h = 24$	$h = 36$	$h = 1$	$h = 12$	$h = 24$	$h = 36$
Panel A: $Macro_{t+1,t+h} = \alpha_h + \beta_h TPU_t + \epsilon_{t+1,t+h}$												
β	-0.12***	-0.13*	-0.14*	-0.13	-0.19***	-0.16**	-0.12*	-0.11*	-0.16***	-0.17**	-0.13	-0.09
NW- t	-3.25	-1.61	-1.35	-1.23	-5.06	-1.98	-1.53	-1.63	-3.23	-1.93	-1.25	-0.84
R^2	1.36	1.68	1.80	1.62	3.74	2.98	2.82	3.48	2.58	3.41	2.57	1.38
Panel B: $Macro_{t+1,t+h} = \alpha_h + \beta_{\text{high},h} S_t^{\text{high}} TPU_t + \beta_{\text{low},h} S_t^{\text{low}} TPU_t + \epsilon_{t+1,t+h}$												
β_{high}	-0.16***	-0.19***	-0.18**	-0.15*	-0.15***	-0.16**	-0.12*	-0.08*	-0.11***	-0.17***	-0.14**	-0.10*
NW- t	-4.71	-2.61	-2.04	-1.53	-4.31	-1.97	-1.59	-1.33	-2.51	-3.14	-2.07	-1.34
β_{low}	0.03	0.03	0.02	0.00	-0.09**	-0.03	-0.03	-0.05	-0.08*	-0.02	-0.01	0.00
NW- t	0.81	0.44	0.23	-0.05	-2.00	-0.42	-0.40	-0.93	-1.37	-0.22	-0.06	-0.04
R^2	2.28	2.84	2.76	2.07	3.74	3.31	3.10	3.48	2.59	3.82	3.05	1.65
R^2_{high}	5.70	6.95	7.92	6.91	7.69	10.70	10.19	8.27	4.25	7.95	6.45	3.91
R^2_{low}	-0.33	-0.38	-0.22	0.11	1.89	0.64	0.70	1.90	1.54	0.70	0.31	0.14

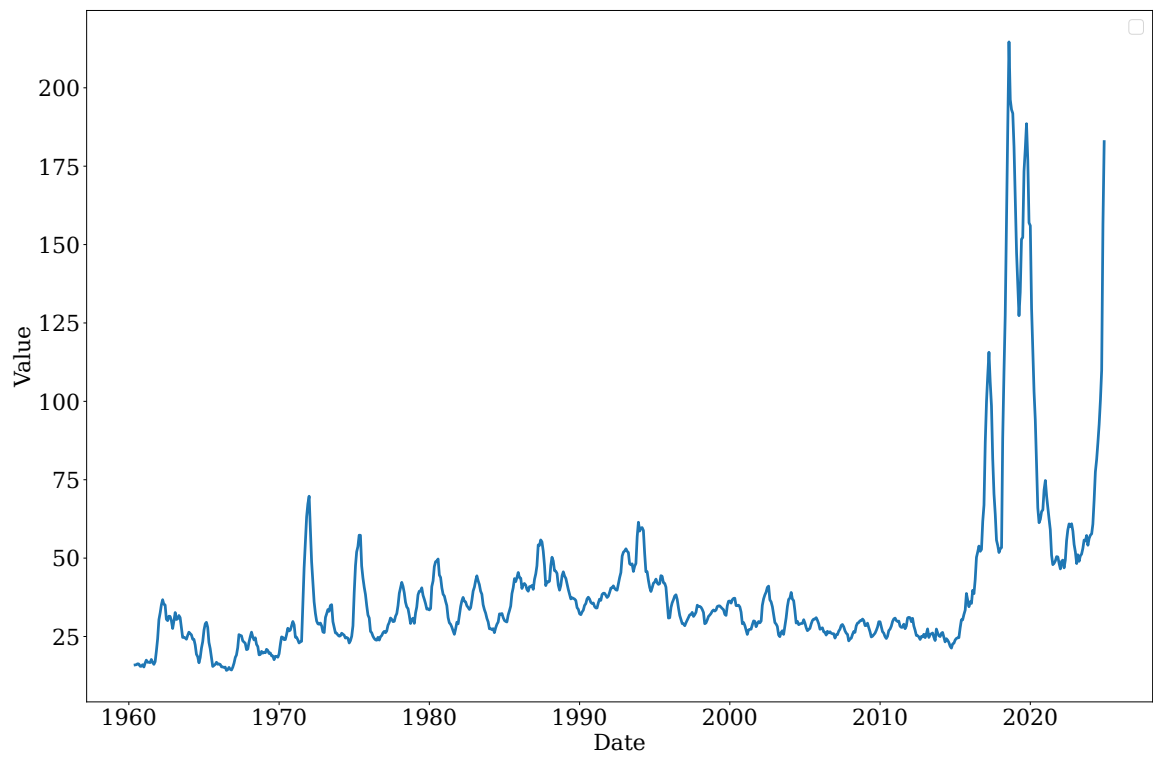


Figure 1: **Trade Policy Uncertainty Index.** This figure illustrates the six-month moving average values of the trade policy uncertainty (TPU) from January 1960 to December 2024. A value of 100 indicates that one percent of all newspaper articles discuss trade policy uncertainty.